

EXP1:

Variable Transformation between Cartesian, Cylindrical and Spherical coordinate system to obtain the location of any object.

TEST CASE 1

Code:

```
clc;

x=input('Enter the value of X=');
y=input('Enter the value of y=');
z=input('Enter the value of z=');

r=sqrt(x^2+y^2);
pi=atand(y/x);

disp('Cartesian to cylindrical coordinate system of p(r,pi,z)=[r pi z]');

r=sqrt(x^2+y^2+z^2);
theta=acosd(z/r);
pi=atand(y/x);

disp('Cartesian to spherical coordinate system of s(r,theta,pi)=[r theta pi]');
```

Calculation:

Rectangular

Case 1: $x = -1$
 $y = 5$
 $z = -8$

Cylindrical

$r = \sqrt{x^2 + y^2} = \sqrt{(-1)^2 + (5)^2} = \sqrt{1 + 25} = \sqrt{26} = 5.099$
 $\phi = \tan^{-1}\left[\frac{y}{x}\right] = \tan^{-1}\left[\frac{5}{-1}\right] = -78.69^\circ$
 $z = -8$

Spherical

$r = \sqrt{x^2 + y^2 + z^2} = \sqrt{(-1)^2 + (5)^2 + (-8)^2} = \sqrt{1 + 25 + 64} = \sqrt{90} = 9.48$
 $\theta = \cos^{-1}\left[\frac{z}{r}\right] = \cos^{-1}\left[\frac{-8}{9.48}\right] = \cos^{-1}(-0.844) = 147.48^\circ$
 $\phi = \tan^{-1}\left[\frac{y}{x}\right] = \tan^{-1}\left[\frac{5}{-1}\right] = -78.69^\circ$

OUTPUT:

Scilab 2025.0.0 Console

File Edit Control Applications ?



Scilab 2025.0.0 Console

Enter the value of X=-1

Enter the value of y=5

Enter the value of z=-8

"Cartesian to cylindrical coordinate system of p(r,pi,z)="
5.0990195 -78.690068 -8.
"Cartesian to spherical coordinate system of s(r,theta,pi)="
9.486833 147.48748 -78.690068

-->

exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci) - SciNotes

File Edit Format Options Window Execute ?



exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci) - SciNotes

exp333.sci X

```
1 clc;-
2 x=input('Enter the value of X=');
3 y=input('Enter the value of y=');
4 z=input('Enter the value of z=');
5 r=sqrt(x^2+y^2);
6 pi=atan2(y,x);
7 disp('Cartesian to cylindrical coordinate system of p(r,pi,z)=[r pi z]');
8 r=sqrt(x^2+y^2+z^2);
9 theta=acosd(z/r);
10 pi=atan2(y,x);
11 disp('Cartesian to spherical coordinate system of s(r,theta,pi)=[r theta pi]');
12
```

TEST CASE 2

Code:

```
clc;
r=input('Enter the value of r=');
pi=input('Enter the value of pi=');
z=input('Enter the value of z=');
x=r*cosd(pi);
y=r*sind(pi);
Z=z;
disp('cylindrical to cartesian coordinate system of p(x,y,z)=[x y z]');
r=sqrt(r^2+z^2);
pi=pi;
theta=acosd(z/r);
disp('cylindrical to spherical coordinate system of s(r,pi,theta)=[r pi theta]');
```

Calculation:

2) Case: 2

$\phi = 30^\circ$

$z = -5$

cylindrical

Rectangular

$x = r \cos \phi = 5 \cos(30^\circ) = 4.33$

$y = r \sin \phi = 5 \sin(30^\circ) = 2.5$

$z = z = -5$

Spherical

$r = \sqrt{\rho^2 + z^2} = \sqrt{(5)^2 + (-5)^2} = \sqrt{25 + 25} = \sqrt{50} = 7.07$

$\phi = \phi = 30^\circ$

$\theta = \tan^{-1} \left[\frac{\rho}{z} \right] = \tan^{-1} \left[\frac{5}{-5} \right] = \tan^{-1}(-1) = \pm 45^\circ$

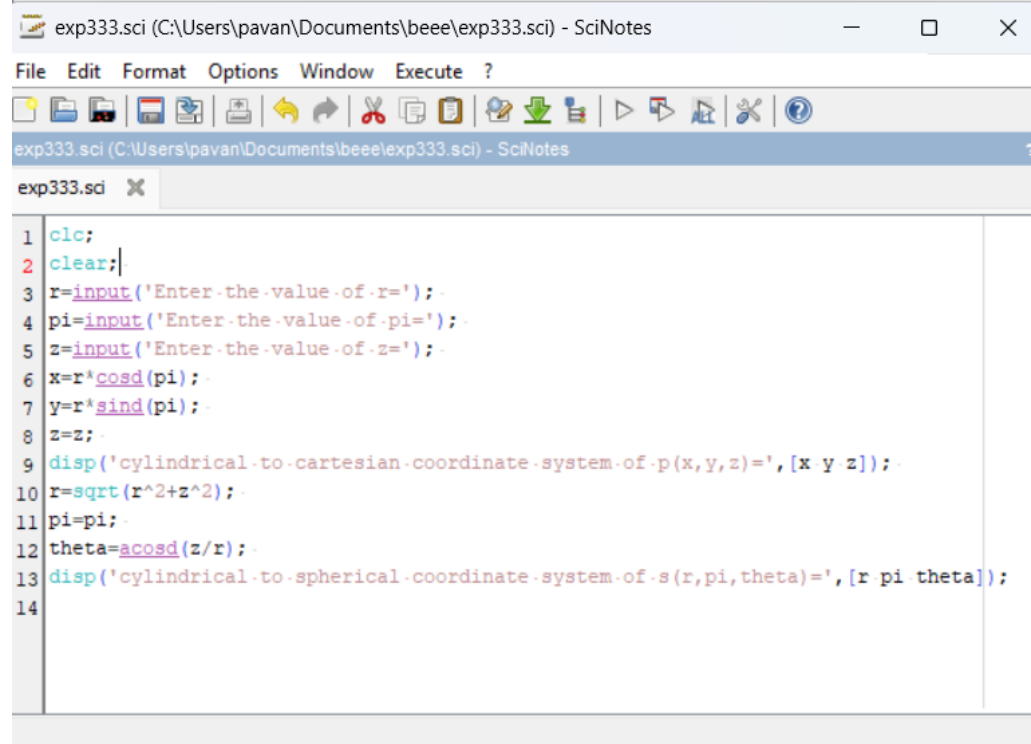
Output:

```
Enter the value of r=5

Enter the value of pi=30

Enter the value of z=-5

"cylindrical to cartesian coordinate system of p(x,y,z)="
4.330127 2.5 -5.
"cylindrical to spherical coordinate system of s(r,pi,theta)="
7.0710678 30. 135.
```



The screenshot shows a SciNotes window titled "exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci) - SciNotes". The window contains the following MATLAB code:

```
1 clc;
2 clear;
3 r=input('Enter the value of r=');
4 pi=input('Enter the value of pi=');
5 z=input('Enter the value of z=');
6 x=r*cosd(pi);
7 y=r*sind(pi);
8 z=z;
9 disp('cylindrical to cartesian coordinate system of p(x,y,z)=[x y z]');
10 r=sqrt(r^2+z^2);
11 pi=pi;
12 theta=acosd(z/r);
13 disp('cylindrical to spherical coordinate system of s(r,pi,theta)=[r pi theta]');
14
```

TEST CASE 3

Code:

```
clc;

r=input('Enter the value of r=');

pi=input('Enter the value of pi=');

theta=input('Enter the value of theta=');

x=r*sind(theta)*cos(pi);

y=r*sind(theta)*sin(pi);

z=r*cosd(theta);

disp('spherical to cartesian coordinate system of p(x,y,z)=[x y z]');

p=r*sin(theta);

pi=pi;

z=r*cos(pi);

disp('spherical to cylindrical coordinate system of s(p,pi,z)=[p pi z]');
```

Calculation:

3) Case: 3

Spherical

$r = 3$

$\theta = 60^\circ$

$\phi = 120^\circ$

Rectangular

$x = r \sin \theta \cos \phi = 3 \sin(60^\circ) \cos(120^\circ) = -2.115$

$y = r \sin \theta \sin \phi = 3 \sin(60^\circ) \sin(120^\circ) = 2.598$

$z = r \cos \theta = 3 \cos(60^\circ) = 1.5$

Cylindrical

$\rho = r \sin \theta = 3 \sin(60^\circ) = 2.598$

$\phi = \phi = 120^\circ$

$z = r \cos \theta = 3 \cos(60^\circ) = 1.5$

Output:

Enter the value of r=3

Enter the value of pi=60

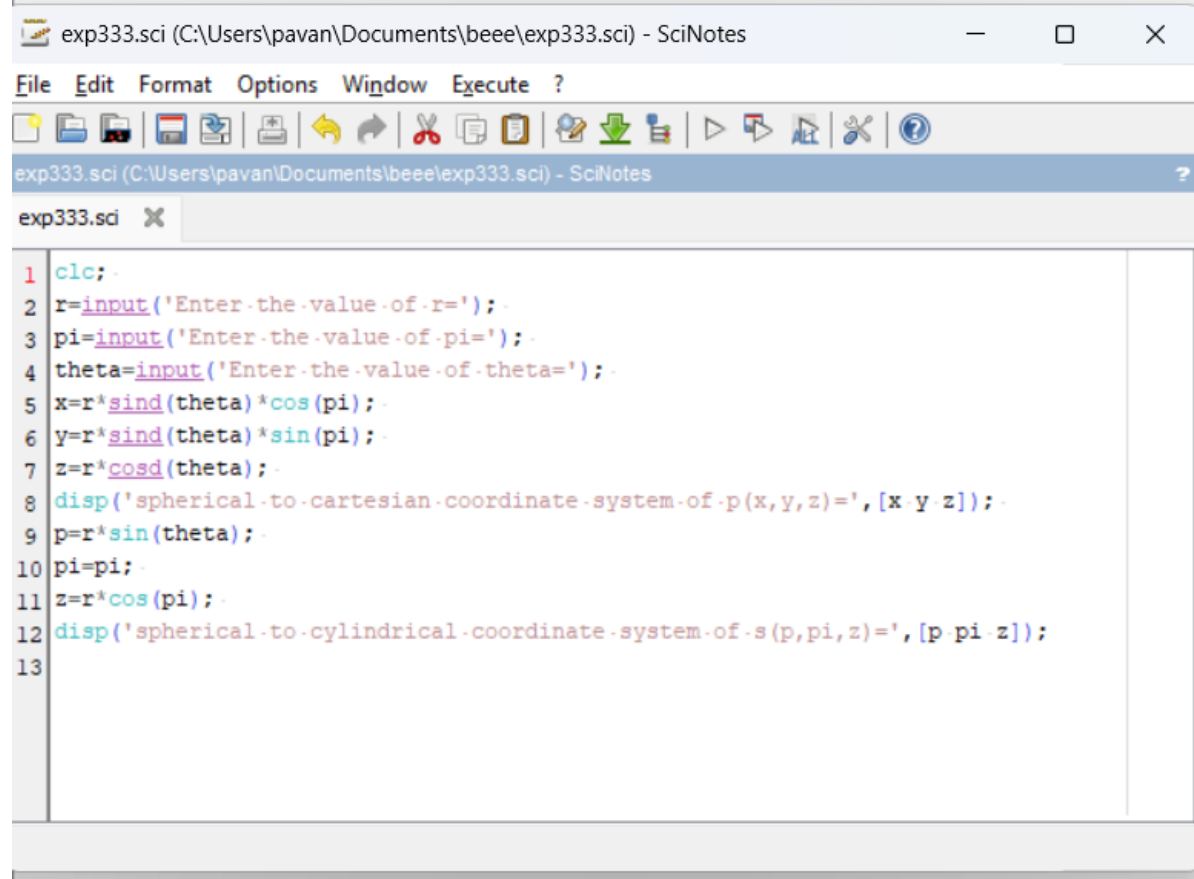
Enter the value of theta=120

"spherical to cartesian coordinate system of p(x,y,z)="

-2.4744415 -0.7919212 -1.5

"spherical to cylindrical coordinate system of s(p,pi,z)="

1.7418336 60. -2.8572389



The screenshot shows a SciNotes window titled "exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci) - SciNotes". The window has a menu bar with "File", "Edit", "Format", "Options", "Window", "Execute", and "?". Below the menu bar is a toolbar with various icons for file operations and execution. The main text area contains the following MATLAB code:

```
1 clc; -
2 r=input('Enter the value of r='); -
3 pi=input('Enter the value of pi='); -
4 theta=input('Enter the value of theta='); -
5 x=r*sind(theta)*cos(pi); -
6 y=r*sind(theta)*sin(pi); -
7 z=r*cosd(theta); -
8 disp('spherical to cartesian coordinate system of p(x,y,z)=[x y z]'); -
9 p=r*sin(theta); -
10 pi=pi; -
11 z=r*cos(pi); -
12 disp('spherical to cylindrical coordinate system of s(p,pi,z)=[p pi z]'); -
13
```